

REU-HPC: Week-4 Group Meeting

Group Meetings

Tuesday 3-4pm (ROC room 1.109)

<https://utexas.zoom.us/j/85978334209>

Week 4 Objectives (June 22 – June 28)

The primary goal for week 4:

- Learn how to install software packages and their dependencies on TACC systems.
- Run sample workloads with and without PEAK to collect profiling data and analyze the usage of mathematical libraries such as BLAS, LAPACK, and FFTW.
- Explored and developed build scripts for additional software packages from the Top 100 applications list by reviewing official documentation, searching for existing examples, and leveraging AI-assisted guidance.

Useful information

- **Frontera-top100-2025-1.txt:**

Link: <https://drive.google.com/file/d/15VXVPnREFcz3hiVyVoCW8ThX5TO6WELO/view?usp=sharing>

- **package_list.xlsx**

Link:

<https://docs.google.com/spreadsheets/d/1JSHFZpOV8IA3M1mw86WJAuWZ8kotStPf/edit?usp=sharing&ouid=101096215360805228479&rtpof=true&sd=true>

- Project directory: **/work2/05392/cylu/share/reu_2026/2.project**

- **App build scripts:** abinit, berkeley_gw, dftb+, epw, gpaw, gromacs, lammmps, lapack, meep, openblas, yambo
- **Examples:** dftb+, gromacs, lammmps, siesta, yambo